

TSFM-Lens: Controlled, Causal and Confirmatory Interpretability for Time-Series Foundation Models

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Abstract

Time-series foundation models (TSFMs) are deployed zero-shot, yet the tools that made language models inspectable do not transfer: tokens span different lengths of time across architectures, no ground truth defines a forecasting “concept”, and causal effects of internal features are small enough that uncontrolled measurements produce confident false positives. We present TSFM-LENS, an open-source system that turns a list of checkpoints into one self-contained interpretability report with one command. It is controlled: a leakage-audited synthetic benchmark with known generative components validates every probe. It is causal: concepts are defined by what removing them does to the forecast, against matched random-direction nulls and reach controls. It is confirmatory: exploratory claims are registered and tested once on a sealed split. Applied to TimesFM, Chronos-2, Sundial and Chronos-Bolt, it finds that causal features mainly change forecast level and shape; that models agree on which inputs a concept groups (20 of 20 registered claims confirmed on 800 held-out series) but rarely on what it does to them (11 of 1,223 tests); and that no concept is shared by all four models. We also document fourteen plausible results that the system’s own controls showed to be wrong.

Keywords: mechanistic interpretability, time-series foundation models, sparse autoencoders, causal intervention, pre-registration

1 Introduction

Pretrained forecasting models are now default components. TimesFM (Das et al., 2024), the Chronos family (Ansari et al., 2024, 2025), Sundial (Liu et al., 2025), Timer (Liu et al., 2024), Moirai (Woo et al., 2024), MOMENT (Goswami et al., 2024), Lag-Llama (Rasul et al., 2023) and Time-MoE (Shi et al., 2025) are released as open checkpoints and used zero-shot. Their architectures differ sharply (decoder-only stacks over 32-step patches, encoder-decoders over quantized single steps, encoders with cross-series attention, sampled flow-matching heads), and leaderboards say which is more accurate. They do not say what these models represent, whether architectures this different represent the same things, or whether any internal structure we can name drives the forecast.

Mechanistic interpretability offers probes (Alain and Bengio, 2017; Hewitt and Liang, 2019), representational similarity (Kornblith et al., 2019), activation patching (Vig et al., 2020; Meng et al., 2022; Zhang and Nanda, 2024), lenses (Belrose et al., 2023) and sparse dictionaries (Bricken et al., 2023; Gao et al., 2025), packaged in libraries such as TransformerLens (Nanda and Bloom, 2022) and pyvene (Wu et al., 2024), and early work applies pieces of this to TSFMs (Wiliński et al., 2025; Mishra, 2026; Oublal et al., 2026; Divo et al., 2026). Three properties of forecasting models break the direct transfer, and each causes silent errors rather than crashes:

- (i) **Tokens are not comparable across models.** One token is 1 time step in Chronos-T5, 32 in TimesFM, 16 in Chronos-2 and 96 in Timer, so every cross-model comparison needs a verified mapping from tokens to time.
- (ii) **There is no ground truth for concepts.** Real series carry no labels for trend, period, changepoints or noise process, so a probe that “finds seasonality” cannot be checked.
- (iii) **Effects are small and easy to fake.** Removing one sparse feature typically moves a forecast by a few hundredths of the context’s standard deviation, less than the dictionary’s own reconstruction error. At that scale a precision mismatch, a patch the head never reads or a null measured on the wrong rows yields a clean, confident and wrong result.

We present TSFM-LENS, a system designed around these problems, and use it to study three questions: **RQ1**, what forecast-relevant concepts TSFMs learn; **RQ2**, whether these concepts are shared across architectures, in the sense of responding to the same inputs, having the same effect, or both; and **RQ3**, whether they causally affect the forecast and how much of it they account for. A supporting question, **RQ4**, asks where in depth these computations happen and how comparable depth is across architectures.

TSFM-LENS rests on three commitments. It is *controlled*: a synthetic benchmark with exact generative ground truth and a battery of structured perturbations gives every probe a known answer. It is *causal*: a feature counts only if removing it changes the forecast more than removing a random direction of the same size on the same series, with reach controls proving the intervention landed. It is *confirmatory*: dev findings are frozen in a hash-pinned registry and tested once on a sealed private split. Measured *time alignment* makes architectures comparable, and a *one-command report* explains every figure in plain language.

Contributions.

- C1.** An open-source system (~61k lines, 171 test modules) that turns a configuration listing checkpoints into a self-contained HTML report through 19 fingerprinted stages, with a zero-code adapter path (Section 4).
- C2.** A controlled benchmark generator with two leakage tiers, a leakage audit, sealed splits with fresh epochs and bit-exact regeneration (Section 4.1).
- C3.** A causal concept pipeline that separates two notions of sharing that are usually conflated: *same inputs* and *same effect on the same inputs* (Section 5.3).
- C4.** A statistical protocol with series-level resampling, a multiplicity ledger and one-shot confirmation (Section 6).
- C5.** A four-model study answering RQ1–RQ4 with typed evidence and held-out replication (Section 7), and a catalogue of fourteen cases in which an uncontrolled analysis gave a plausible, wrong answer that a built-in control caught (Section 8).

Main findings. On TimesFM 2.5, Chronos-2, Sundial and Chronos-Bolt: (RQ1) causal sparse features exist (293 across 29 layers clear a row-matched null) but act mostly on forecast *level* and *shape*, and tile a continuum rather than forming discrete concepts; (RQ2)

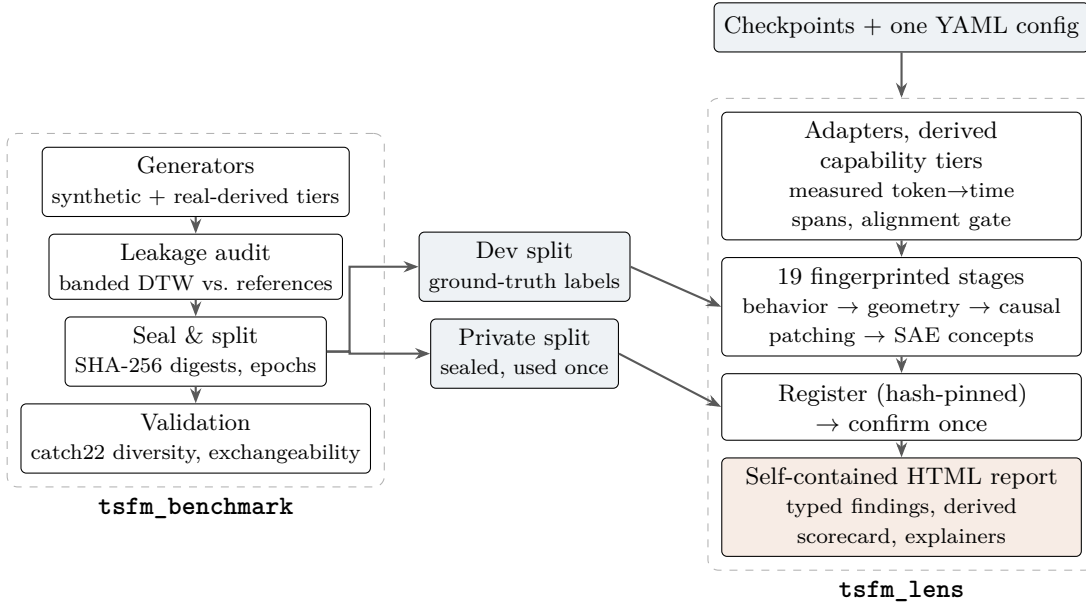


Figure 1: System overview. **tsfm_benchmark** produces leakage-audited, sealed corpora with exact ground truth; **tsfm_lens** turns a configuration naming checkpoints into one report. Dev findings are exploratory; the private split is opened once, by **confirm**, for claims registered in advance.

models agree on *which inputs* a concept groups (20 of 20 registered transfer claims confirmed on 800 sealed series) but rarely on *what it does* to them (11 of 1,223 shared-input tests show the same causal effect), and no concept is shared by all four models; (RQ3) single features are causal but small, and 64% of visually prominent features have no causal effect; (RQ4) depth-resolved reaction profiles replicate held-out (9 of 9 corruptions), but depth locations are comparable only after accounting for captured compute (14% to 98% of forecast FLOPs).

The full ledger of results, each with exact numbers, evidence class, status and a reproduction pointer, is **FINDINGS.md** (indexed in Appendix A). The example run of Section 7, with its report and every per-stage artifact, is **examples/concept_atlas_v2**; **paper/figures/make_figures.py** regenerates every data figure from it.

2 Related Work

TSFMs. TimesFM (Das et al., 2024) is decoder-only over 32-step patches; Chronos (Ansari et al., 2024) quantizes values and trains a T5 encoder–decoder (Raffel et al., 2020) with a sampled decoder; Chronos-Bolt uses a direct patch head, and Chronos-2 (Ansari et al., 2025) is an encoder with cross-series group attention; Sundial (Liu et al., 2025) samples from a flow-matching head (Lipman et al., 2023); Timer (Liu et al., 2024) uses 96-step tokens and Lag-Llama (Rasul et al., 2023) reads lagged values. Most normalize each series internally (Kim et al., 2022), a detail that matters when an adapter calls a model below its public API (Section 8). Wan et al. (2026) document flat “collapsed” forecasts on weakly predictable targets, which we also measure (PM-07).

Interpreting time-series models. Saliency methods for time-series classifiers are unreliable across architectures (Ismail et al., 2020); Kalnāre et al. (2025) apply patching and SAEs to time-series classifiers. For TSFMs, Wiliński et al. (2025) study representational redundancy and concept steering, and four recent studies train SAEs on one TSFM (Mishra, 2026; Oublal et al., 2026; Divo et al., 2026; Yıldırım, 2026). Each examines one model or one method. None compares architectures on a common time axis, validates probes against generative ground truth, separates input agreement from effect agreement, or tests findings on held-out data.

Methods, tooling and validity. Probes need control tasks (Hewitt and Liang, 2019; Belinkov, 2022); patching is sensitive to metrics and corruptions (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024) and can create illusions (Makelov et al., 2024); attention weights are not explanations (Jain and Wallace, 2019); saliency maps can fail sanity checks (Adebayo et al., 2018). SAE features vary across seeds (Paulo and Belrose, 2025), and crosscoders learn features jointly across models (Lindsey et al., 2024). Representation similarity (Raghu et al., 2017; Kornblith et al., 2019; Bansal et al., 2021) underlies claims of convergence (Li et al., 2016; Huh et al., 2024). Libraries such as TransformerLens, nnsight (Fiotto-Kaufman et al., 2025), pyvene, SAELens (Bloom et al., 2024) and Captum (Kokhlikyan et al., 2020) provide primitives for one model; they do not provide comparative designs, nulls, multiplicity control or confirmation (Table 1). From empirical science we import pre-registration (Nosek et al., 2018), protection against adaptive reuse of holdouts (Dwork et al., 2015), leakage auditing (Kapoor and Narayanan, 2023) and resampling at the level of the independent unit (Efron and Tibshirani, 1993; Cameron et al., 2008).

3 Design Principles

Each principle corresponds to a failure that actually happened (Section 8) and is enforced in code as a gate, refusal or test.

- P1. Ground truth by construction.** Every probe is first validated on a case whose answer is known: the benchmark records every generative component of every synthetic series, and every perturbation is applied to series whose components are known.
- P2. Alignment is measured, not declared.** Token spans are measured from impulse responses; cross-model analysis happens only on a common window axis; models that are not time-localized receive behavioral analysis only.
- P3. Causal claims need a null, a reach control and a same-seed baseline.** Every intervention is compared with a random intervention of matched size on the same series; self-patching must change the forecast by exactly 0.0 and cross-patching by a nonzero amount; sampled models are seeded identically in compared calls.
- P4. The series is the unit of inference.** All intervals and p -values resample series; p -values are floored at $1/n_{\text{boot}}$, and each correction family’s attainable minimum is checked.
- P5. Explore, then confirm once.** Dev results are exploratory; registered claims are tested once on a sealed split, and a second look is recorded.
- P6. Every claim is typed; failures are loud.** Each finding carries an evidence class (BEHAVIORAL, DESCRIPTIVE, GEOMETRIC, LINEAR-TRANSLATABLE, CAUSAL WITHIN-

Capability	<i>TransformerLens</i>	<i>msight / pyvene</i>	<i>SAELens</i>	<i>Captum</i>	<i>TSFM SAE studies[†]</i>	<i>TSFM-Lens</i>
Hooks, caching, activation patching	✓	✓	○	○	○	✓
Sparse dictionaries and feature ablation	○	○	✓	✗	✓	✓
Time-series models without per-model engineering	✗	○	✗	○	✗	✓
Measured token→time alignment across architectures	✗	✗	✗	✗	✗	✓
Benchmark with exact generative ground truth, leakage audit	✗	✗	✗	✗	○	✓
Random-direction nulls and reach controls for every ablation	✗	✗	✗	✗	✗	✓
Cross-model concept matching (inputs vs. effects)	✗	✗	✗	✗	✗	✓
Series-level resampling and multiplicity ledger	✗	✗	✗	✗	✗	✓
Registered, one-shot confirmation on sealed data	✗	✗	✗	✗	✗	✓
Typed evidence class and generated caveat per claim	✗	✗	✗	✗	✗	✓
One-command, novice-readable report	✗	✗	○	✗	✗	✓

Table 1: Capabilities for comparative TSFM interpretability. ✓ provided; ○ possible with user code or partial; ✗ not provided. General libraries are single-model microscopes and complementary to TSFM-LENS, which uses the same primitives. [†]Mishra (2026); Oublal et al. (2026); Divo et al. (2026); Yıldırım (2026), judged from the papers.

MODEL, ILLUSTRATIVE, HELD-OUT CONFIRMED) and a generated caveat; unsupported analyses are skipped with a rendered reason.

P7. Accessible by default. The deliverable is one HTML file a non-specialist can read, with every figure explained and every verdict derived from artifacts rather than written by hand.

New behavior is opt-in so sealed corpora regenerate bit-exactly, and stage configurations are fingerprinted so stale cached artifacts are refused.

4 System Architecture

The repository holds two separately installable packages: `tsfm_benchmark` (one subpackage for generation, audit and sealing, another for diversity and exchangeability) and `tsfm_lens` (subpackages for models, extraction, analysis, SAEs and the report, plus the stage orchestrator `pipeline.py`). The folder `examples/concept_atlas_v2` contains the report, configuration and every lightweight artifact of the run analyzed in Section 7, so the report and this paper’s figures can be regenerated without GPUs. The `main` branch is the clean publishable surface; a `dev` branch keeps the research log, standalone study drivers and experiment configurations.

4.1 Controlled benchmark generation

Real data enter the benchmark only as leakage references, as realism calibration and as raw material for explicitly labeled generators. The *synthetic* tier touches no real data and has

exact ground truth: **parametric** is additive (trend, multiple seasonalities, AR noise, change-points, anomalies, random walk, intermittency, heteroskedasticity), and **random_parametric** draws a recipe from one of eight frozen archetypes (four more are opt-in). The *real-derived* tier (**mixture**, **block_bootstrap** (Künsch, 1989), **sequential_par**) inherits its source distribution and has attributional ground truth. Because “no leakage” and “realistic” pull in opposite directions, each sample carries its tier label and analyses report per tier.

Every sample passes a two-stage leakage gate against the Monash archive (Godaheva et al., 2021) (a cheap prefilter, then banded DTW; Sakoe and Chiba, 1978) and a cross-split near-duplicate check. Dev and private splits use disjoint seed ranges mixed with SHA-256; every sample is hashed so that `load_sealed(verify=True)` detects tampering, and fresh private *epochs* are minted by striding the seed space. A separate validation package measures redundancy by shape matching and diversity in the catch22 feature space (Lubba et al., 2019) (participation-ratio dimension, Gao et al., 2017; nearest-neighbor tails), never on a 2-D embedding, and tests dev/private exchangeability with an energy distance (Székely and Rizzo, 2013) plus TOST equivalence (Schuirmann, 1987) (v1 benchmark: energy $p = 0.51$, 21 of 22 features equivalent; MP-09).

Two perturbation families are built in: nine structured *corruptions* (noise, detrend, deseasonalize, frequency shift, level shift, spike, smooth, warp, dropout), which drive L3, and *generator counterfactuals*, which move one generative knob along a dose ladder with every random draw fixed. Dose 1.0 must reproduce the original bit-exactly, a free identity test that caught a real bug (BM-05).

4.2 Model adapters and capability tiers

A model enters through an adapter (`tsfm_lens/tsfm_lens/models/base.py`) exposing `load`, `predict` (point forecast and quantiles in the original scale), the module, input preparation, a forward pass and `token_time_spans`. The *capability tier* is derived from which methods a subclass overrides, never declared: 0 (black box), 1 (observable hidden states with spans), 2 (steerable: single-pass context, so patching is possible) and 3 (decomposable attention). Stages above a model’s tier are skipped with the missing method named in the report. A missing capability (`CapabilityUnavailable`) is a fact about the adapter; a measured failure of time localization (`NotTimeLocalized`) is a fact about the model and routes it to behavioral analysis only.

New models use the zero-code **generic_hf** adapter, which discovers layers and measures spans and refuses rather than guesses (verified on Timer, whose 96-step spans it recovers from measurement alone; MP-01), or a copy of `tsfm_lens/tsfm_lens/models/TEMPLATE_adapter.py` in `models/contrib/`. `run.py --check-adapter` then runs patch-identity and reach conformance, the alignment gate, span discovery and capability checks. Table 2 lists the models.

4.3 Time alignment across architectures

Each token’s span is measured from its impulse response (`extraction/span_discovery.py`), gated on peak-to-pedestal contrast and contiguity. Given spans I_j and windows $W_w =$

Model	Checkpoint	Architecture	Token	Head	Tier	Params	FLOPs obs.
TimesFM 2.5	google/timesfm-2.5-200m	decoder-only	32	deterministic	3	231M	42.5%
Chronos-2	amazon/chronos-2	encoder (+group attn.)	16	deterministic	3	119M	97.5%
Sundial	thuml/sundial-base-128m	decoder + flow head	16	sampled	2	128M	22.4%
Chronos-Bolt	amazon/chronos-bolt-small	encoder + patch head	16	deterministic	2	48M	79.3%
Chronos-T5	amazon/chronos-t5-*	encoder-decoder	1	sampled	3	8M–709M	14.4% ^a
Timer	thuml/timer-base-84m	decoder-only	96	deterministic	2 ^b	84M	–
Lag-Llama	Lag-Llama	decoder over lags	lags	sampled	2	–	–

Table 2: Models with adapters; the first four form the case study. “Token” is time steps per token; “FLOPs obs.” is the share of a forecast’s FLOPs spent in captured blocks, measured with PyTorch’s FLOP counter. ^aChronos-T5-Base, encoder only (DE-05). ^bVia `generic_hf`.

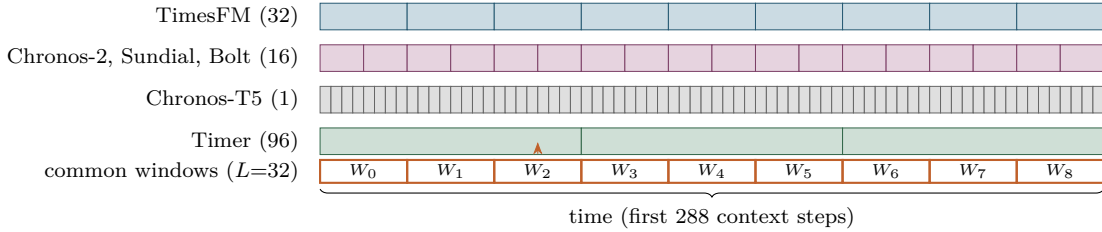


Figure 2: Time alignment. Tokens cover measured spans of different widths (Chronos-T5’s one-step tokens drawn in groups of four); Equation (1) pools them onto common windows.

$[wL, (w+1)L)$ ($L = 32$ by default), tokens are overlap-pooled,

$$P_{wj} = \frac{|I_j \cap W_w|}{\sum_{j'} |I_{j'} \cap W_w|}, \quad \tilde{h}_w = \sum_j P_{wj} h_j, \quad (1)$$

and every cross-model analysis runs on [series, window, dim] tensors (Figure 2); an empty window raises rather than yielding a zero row. During extraction an *alignment gate* injects an impulse into each window and checks, per layer, that the matching pooled window moves most, normalizing hits by their attainable ceiling (a 96-step token on 32-step windows reaches at most one third; MP-02), with the impulse amplitude self-calibrated per checkpoint and context length.

4.4 The stage pipeline and its gates

An analysis is a pipeline of 19 stages (Figure 3) writing typed artifacts. Stages skip themselves when their artifacts exist; each declares the configuration keys it depends on, which are fingerprinted so that a changed setting marks exactly the dependent artifacts stale. Three gates write their decisions to JSON and the report names them as skip reasons: the *tier* gate, the *run-shape* gate (comparison stages are dropped for one model) and the *routing* gate. Extraction captures the residual stream into a float16 zarr store, and every intervention goes through one primitive (`hooks.token_patch`), which raises if the model silently splits a batch.

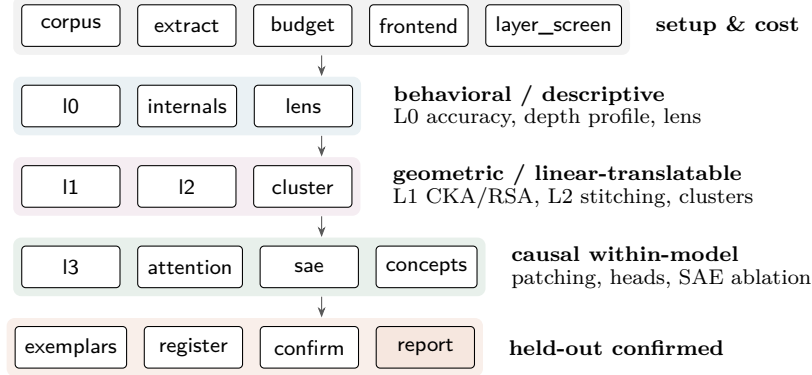


Figure 3: The 19-stage pipeline, grouped by the strongest evidence class each group can produce. Gates (tier, run shape, routing, configuration fingerprints) skip a stage per model with a rendered reason; only **confirm**, on the sealed private split, produces HELD-OUT CONFIRMED evidence.

4.5 The report: interpretability for non-specialists

The deliverable is one self-contained HTML file (`examples/concept_atlas_v2/report.html`), written so that a reader without interpretability training can follow it and an expert can audit every number (Figure 4).

- **Question-first.** Seven parts pose plain questions (“Is the comparison fair?”, “Where does the forecast form?”, “What causally drives the forecast?”, ...); each stage opens with a card stating its question, method, what good and bad results look like, and what it cannot tell you.
- **Every figure explained.** A caption plus a collapsible “What does this mean?” note (purpose, how to read, limitations): 361 notes in the example. A glossary and methods appendix are generated from the code, and a toggle switches between headline, standard and methods views.
- **Derived verdicts.** The scorecard (33 rows) is a pure reduction over artifacts with no model names hard-coded; each row shows the value, its reference (null, floor, ceiling or baseline) and the rule that yields the verdict.
- **Typed findings.** Each claim is a **Finding** record with evidence class, plain-language sentence and generated caveat: 126 in the example (58 descriptive, 29 behavioral, 21 causal, 12 linear-translatable, 4 geometric, 2 illustrative).
- **Nothing silently missing.** Skipped stages render their gate reason, failed sections render visibly, and a multiplicity ledger (45 families, 3,599 comparisons) flags families that cannot reach α .

The complete analysis is one command,

```
python run.py --config configs/examples/large_4model.yaml
```

and smaller presets cover a CPU-only mock smoke test, a single model and a pair.

L3

Perturbation & patching

Where each model's depth reacts to structured corruptions, and where clean activations causally restore corrupted forecasts.

What this stage tells you

QUESTION

Where, causally, does each model actually carry a specific structural property of the data — and does damaging it there hurt the forecast?

HOW

Corrupts one structural property of the input at a time (trend, seasonality, added noise, and so on), measures how much each layer's activations shift because of it, then splices the clean, uncorrupted activations back into an otherwise-corrupted forward pass, one layer at a time, to see how much of the clean forecast that single splice restores.

GOOD VS. BAD RESULT

Good: a restoration curve that rises toward 1.0 at some depth for a given corruption — that layer causally carries the fix for that property — and two models whose sensitivity fingerprints peak at matching relative depths for the same corruption. Bad: a restoration curve that never clears a small fraction anywhere, or a corruption whose measured sensitivity is small only because it touches a tiny fraction of the input's timesteps in the first place, not because the model is robust to it.

WHAT IT CANNOT TELL YOU

The activation-shift fingerprints are a magnitude-of-change measure by themselves, not a causal one — a layer can shift a lot without that shift ever affecting the final forecast, which is exactly why the patching curve exists alongside it. Even the patching curve is causal only *within* one model; this stage never transplants activations between models, so any cross-model comparison here is two separately-measured within-model curves side by side, never a joint causal test.

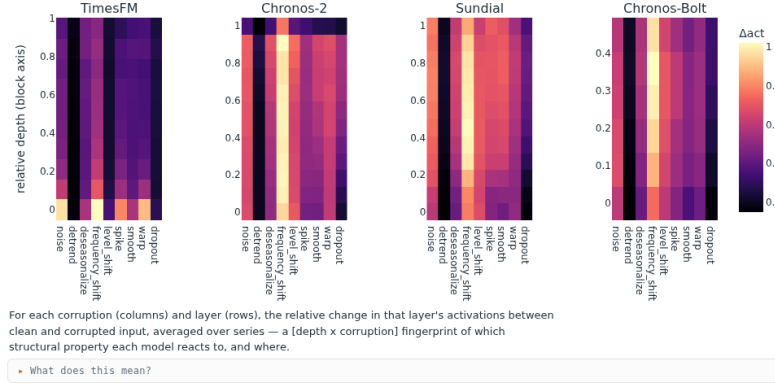


Figure 4: Excerpt of the generated report (L3 section of the example run): the stage card states the question, method, how to read good and bad results and the limitation, above the per-model corruption-sensitivity fingerprints. Every figure carries a caption and an expandable explanation.

5 Measurement Methods

Throughout, $x \in \mathbb{R}^T$ is a context ($T = 512$ in the case study), $y \in \mathbb{R}^H$ its continuation ($H = 64$), $\hat{y}$ a point forecast and $h^{(\ell)}$ the residual stream after block ℓ .

5.1 Behavior, geometry and perturbation (L0–L3)

L0. Accuracy is MASE (Hyndman and Koehler, 2006) per series, with the seasonal lag from ground truth and a scale floor on the naive denominator, without which intermittent series spuriously improve (MN-23); sMAPE, pinball loss (Gneiting and Raftery, 2007) and calibration are reported alongside. Pairs are compared by a paired series bootstrap per family, Holm-corrected across all pairs and families (Holm, 1979), and read against each model's repeat-run noise floor. **budget** measures parameters, FLOPs, latency and captured-FLOP

share; **frontend** tests scale equivariance, NaN handling and quantization before any layer runs.

L1, L2. For aligned, window-pooled and centered activations A, B over the same series, linear CKA (Kornblith et al., 2019) $\|B^\top A\|_F^2 / (\|A^\top A\|_F \|B^\top B\|_F)$ is computed for every layer pair of every model pair, with a cluster bootstrap and a shuffled-series null. L2 fits ridge maps between layers and reports only the *gain* over an input-feature baseline, since much of any layer is linearly predictable from the input. Cluster agreement uses AMI (Vinh et al., 2010). These claims are GEOMETRIC or LINEAR-TRANSLATABLE.

L3. For each corruption c , the sensitivity *fingerprint* is the relative activation change $\|h^{(\ell)}(c(x)) - h^{(\ell)}(x)\| / \|h^{(\ell)}(x)\|$ per layer, and the *restoration curve* is the fraction of forecast damage undone by patching clean activations of layer ℓ (whole, or per window) into the corrupted pass. Models are compared by rank-correlating fingerprints on shared relative depth, never by transplanting activations. Restoration is CAUSAL WITHIN-MODEL; cross-model agreement is correlational.

5.2 Depth

A skip lens decodes the forecast from each layer through the model’s head and a tuned ridge lens (Belrose et al., 2023) fits per-layer readouts; **internals** reports effective dimension, family decodability and CKA to the input. The depth axis is the fraction of the *whole* stack, including uncaptured parts, so Chronos-T5’s last captured block sits at 0.478 (DE-04); and the lens is withheld for a model whose head reads positions the patch never writes, where it would return a flat, meaningless curve (DE-03).

5.3 Causal concepts

Sparse dictionaries. Each target layer gets a TopK SAE (Makhzani and Frey, 2013; Gao et al., 2025), $z(h) = \text{TopK}(W_e(h - b_d) + b_e)$, $\hat{h} = W_d z(h) + b_d$, with searched dictionary size, an auxiliary loss reviving dead atoms and a minimum step count (MP-03). Since single seeds are not reproducible (Paulo and Belrose, 2025), replicate dictionaries give a within-model reproducibility ceiling (MP-04).

The ablation battery. Feature i is tested on the k series where it fires most, by comparing the forecast under the reconstruction with $z_i = 0$ against the full reconstruction:

$$\Delta_i^{(c)}(x) = g_c(f(x; h^{(\ell)} \leftarrow \hat{h}_{-i})) - g_c(f(x; h^{(\ell)} \leftarrow \hat{h})), \quad (2)$$

where g_c is one of nine forecast channels (trend, seasonal strength, spectral centroid, level, dispersion, near/far-horizon shape distance, MASE, flatness). Using the reconstruction as baseline keeps the SAE’s error from being charged to the feature. The null removes a random code direction of matched magnitude on the *same* rows; a feature clears channel c if mean $|\Delta_i^{(c)}|$ exceeds the null’s 95th percentile $q_{95}^{(c)}$, and its channel vector is $v_i^{(c)} = \overline{\Delta_i^{(c)}} / q_{95}^{(c)}$. Shape channels are distances and are treated as undirected. A *reach probe* first requires self-patching to change the forecast by exactly 0.0 and cross-layer patching by a relative $> 10^{-3}$; otherwise the target is withheld with its reason (Figure 5).

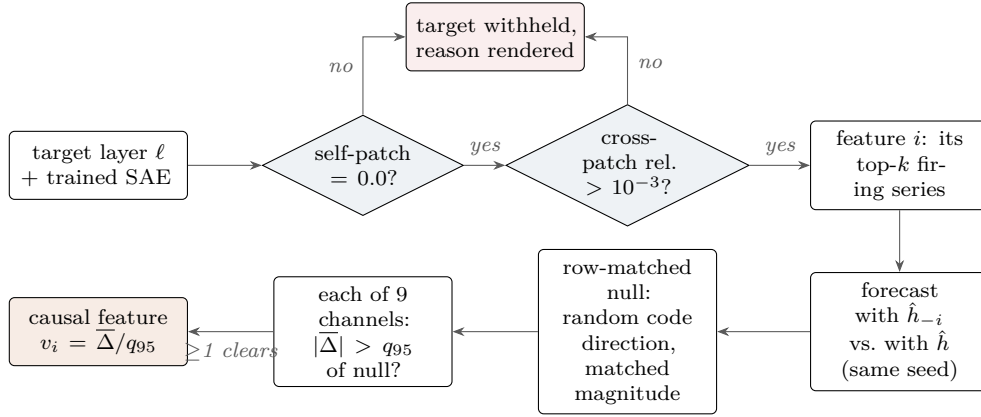


Figure 5: The ablation battery. A target is scored only after both reach controls pass; each feature is compared, channel by channel, with a random-direction null on exactly its own top-firing series.

From features to concepts. Causal features are grouped at three granularities. *Per-target concepts* cluster one layer’s vectors with k -means, choosing k by silhouette (Rousseeuw, 1987) among values whose smallest cluster has ≥ 3 members (otherwise silhouette rewards splitting off an outlier; SH-20); a target with no admissible k is *non-modular*. The *atlas* pools features from all models and layers and cuts a complete-linkage dendrogram at cosine 0.9, so an atlas concept is a shared causal-effect profile. *Families* are a coarser readable tiling (average linkage, cosine 0.5), tested against a column-shuffle null and titled deterministically from their channels.

5.4 Two notions of sharing

“Shared” is two questions, answered separately. *Same inputs*: concept transfer asks whether a feature in another model separates a concept’s top- k series from the rest (AUC), against stratum-matched random subsets, in both directions (*reciprocal* if both clear), with Benjamini-Hochberg (Benjamini and Hochberg, 1995) per ordered model pair and leg. Input agreement between concept parts is top- k overlap (hypergeometric), never whole-series correlation: two sparse features can have $\rho = 0.80$ with no shared top series (MN-19). *Same effect on the same inputs*: every reciprocal-FDR transfer is ablated on both sides on the same series set U ; each side must clear its own null on U , and level concordance and shape cosine must beat both sides’ activation-matched random-feature floors. Falling below the floors’ 5th percentile is “acts differently”; merely failing the 95th is “no specific agreement” (MN-18).

The report combines these into a six-rung ladder per concept (`report/derived.py`): L1 same forecast effect (atlas), L2 same inputs, L3 reproducible across SAE seeds, L4 other dictionaries select the same inputs, L5 same causal effect on the same inputs, L6 confirmed on private data. Each atlas concept also gets a `sharing_class`: *shared* (same effect and inputs), *partially shared*, *convergent* (same effect, different inputs) or *single-model*; a part is provenance-driven when $\geq 80\%$ of its top series come from one real-derived generator.

6 Statistical Protocol

Four mechanisms control how often a report with thousands of comparisons is fooled (Table 3).

The series is the resampling unit. Windows of a series share its trend, noise and scale, so every interval and p -value resamples whole series (Efron and Tibshirani, 1993; Cameron et al., 2008): paired bootstrap for accuracy, cluster bootstrap for CKA, validation-series intervals for stitching, paired cluster bootstrap for patching. Splits are by series and stratified, never head slices of a task-grouped corpus.

A multiplicity ledger. Every test belongs to a named family (Holm for confirmatory families, BH for discovery-oriented atlas transfers). Resampled p -values are floored at $1/n_{\text{boot}}$ (Phipson and Smyth, 2010), so a Holm family of m tests cannot go below m/n_{boot} ; the ledger computes this minimum per family and flags families that can never be significant, a failure that grows quadratically with the number of models. The example has 45 families and 3,599 comparisons; its strictest L0 family (30 tests) has a minimum Holm p of 0.015 at $n_{\text{boot}} = 2000$.

References from the run itself. Effects are read against repeat-run noise floors, shuffled-series nulls, input-feature baselines, random-direction nulls, activation-matched floors and replicate-seed ceilings. Before the private split is opened, `confirm` simulates each claim’s minimum detectable effect (e.g. ΔMASE 0.155 at $n = 240$, 80% power), so a non-replication reads as “absent” or “underpowered”.

Register, then confirm once. `register` freezes hypotheses from dev results together with the content hash of each source artifact, and hashes the registry. `confirm` refuses mismatched hashes, requires the private seal to verify, tests each claim once and refuses to overwrite its artifact; a forced rerun is recorded as `repeated_look`, and a consumed split is replaced by a fresh epoch. This is pre-registration (Nosek et al., 2018) in software, guarding against adaptive holdout reuse (Dwork et al., 2015); unregistered findings cannot be rescued.

7 Case Study: Four Time-Series Foundation Models

7.1 Setup

We analyze TimesFM 2.5, Chronos-2, Sundial and Chronos-Bolt with the preset `tsfm_lens/configs/concept_atlas_v2.yaml`. The dev corpus has 965 series in five generator families; the real-derived ones draw on Monash weather, ETT (Zhou et al., 2021) and the Chronos electricity and traffic collections. Contexts are 512 steps, the horizon 64, the window 32. SAEs are trained on 29 target layers (TimesFM 10, Chronos-2 7, Sundial 9, Chronos-Bolt 3) with three replicate seeds each, and $n_{\text{boot}} = 2000$. The private split is 800 sealed series, opened once. The run took 6.43 hours on one RTX A5000 (24 GB). Artifacts are in `examples/concept_atlas_v2/run`; results from other runs carry their FINDINGS.md identifier.

Table 4 gives the models’ behavior. Chronos-2 is most accurate on dev and held-out data; Sundial’s nominal 80% interval covers 37% of outcomes; TimesFM and Sundial propagate an injected NaN silently. The Sundial row hides a bug found in this work: the first adapter

Claim type	Test statistic	Null / reference	Correction	Held-out
Model A more accurate (per family)	paired Δ MASE, series bootstrap	zero difference; repeat-run noise floor	Holm over all pairs $\times$ families	✓
Layers share geometry	linear CKA, cluster bootstrap	shuffled-series null at the same layer pair	per model pair	✓
Layer linearly translatable	ridge stitching gain	input-feature baseline (gain = 0)	per pair	–
Depth reaction agrees	Spearman ρ of fingerprints on shared depth	bootstrap CI of ρ	per corruption	✓
Layer restores forecast	restoration fraction	self-patch = 0, cross-patch $\neq$ 0	per model	–
Feature is causal	mean $ \Delta^{(c)} $ on own rows	row-matched random-direction q_{95}	per channel (count vs. chance)	–
Concept reproducible	best-match cosine across seeds	within-model replicate ceiling	–	–
Same inputs	top- k AUC in other model	stratum-matched random subsets	BH per ordered pair and leg; Holm when registered	✓
Same effect on same inputs	level concordance; shape cosine	activation-matched feature-set floors (q_{95} agree, q_{05} differ)	per test	–
Families are real	number and coverage of families	column-shuffle null	–	–

Table 3: Every claim type the case study reports, the statistic that tests it, the reference it must beat, its multiplicity correction and whether the **confirm** stage currently re-tests it on private data. Claim types without a held-out mark are reported as exploratory.

Model	median MASE	80% coverage	scale resid. ($\times 0.001$)	NaN input	latency
Chronos-2	1.1251	0.772	0.0093	handled	21 ms
TimesFM	1.1829	0.795	2.0×10^{-6}	propagates	38 ms
Sundial	1.3335	0.369	0.0332	propagates	7 ms
Chronos-Bolt	1.4913	0.749	0.0087	handled	14 ms
naive / seasonal naive	2.6829 / 2.0280				

Table 4: Behavior on the v2 dev corpus (965 series). Scale residual: error of $f(cx)/c$ against $f(x)$ in context standard deviations. Held-out, Chronos-2 beats TimesFM overall (paired Δ MASE -0.148 $[-0.211, -0.088]$, $p = 0.0005$, $n = 800$).

bypassed the checkpoint’s input normalization, and a scale-equivariance residual of 38.43 context standard deviations was at first read as a model property (PM-15, Appendix D; after the fix, 0.033).

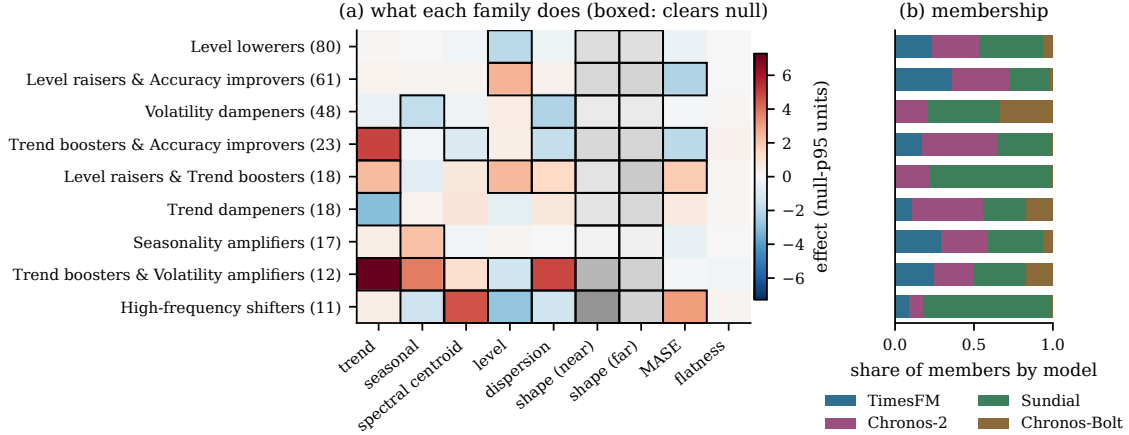


Figure 6: The nine causal-effect families. (a) Family centroid in null units, signed as what the feature *does* (red raises, blue lowers; boxed cells clear the null; shape distances unsigned, gray). (b) Membership by model. Source: `sae/concept_families.json`.

7.2 RQ1: What forecast-relevant concepts do TSFMs learn?

Causal features exist, and the battery discriminates. 293 SAE features across 29 targets clear the row-matched null on at least one channel. Twenty-eight targets passed both reach controls (e.g. TimesFM `stacked_xf.10`: self-patch 0.0, cross-patch reach 0.146). Chronos-2’s final block was withheld: cross-patching moved the forecast by exactly 0, because its head reads placeholder tokens a context patch never reaches (DE-03); without the probe it would have reported a table of zero effects. On the reference run, (feature, channel) cells clear the null at 2.49–6.61 times the chance rate, and randomly nominated features clear at about half the rate of activation-nominated ones (MP-07).

The concepts are about level and shape. Nine effect families tile the causal features (Figure 6). The largest act on level and volatility: *Level lowerers* (80), *Level raisers* & *Accuracy improvers* (61), *Volatility dampeners* (48). Trend and seasonality families are smaller (23, 18, 17 and 12 members), and nearly every family also moves the shape distances. Level is not everything: the median pure-level share of a causal effect is 0.535, and most features still clear a null after the level shift is removed from both feature and null (CA-07).

The concepts are graded, not discrete. Within a layer, 24 of 29 targets are non-modular (9 per-target concepts in total); the strict atlas assigns only 86 of 293 features to its 27 concepts; and the families cover 288 of 293 features but do *not* beat a column-shuffle null ($p = 1.0$ for their number, 0.985 for coverage). On the reference run silhouette is 0.31–0.35 for every family count from 3 to 16 (CA-10) and the ablation fingerprints have participation ratio 2.09 (MN-07). Families are thus a reading vocabulary, not discovered units.

Ground-truth concepts are rare and need a null. After dead-feature revival, 154 of 10,240 TimesFM and 115 of 6,144 Chronos-T5-Base features match a ground-truth field beyond a permutation null (best: intermittency, $\rho = 0.881$; MP-06); raw correlations do not separate trained from untrained weights (0.567 vs. 0.565). Validating concepts by moving a

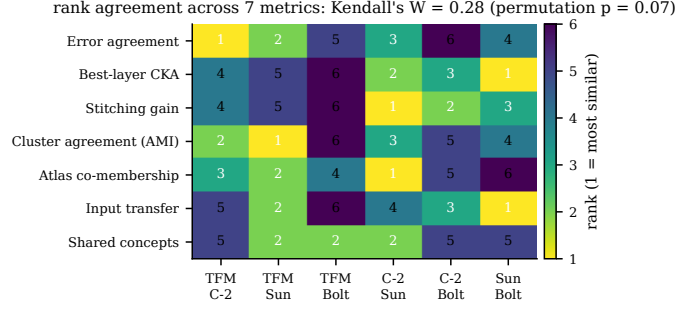


Figure 7: Rank of each model pair (columns) under seven similarity metrics (rows); 1 is most similar. Source: `report/model_similarity.json`.

generator knob was a confirmed negative twice (0 of 98, then 0 of 151 pairs), because a knob moves any active feature about as much as the targeted one (MN-17), and a vivid TimesFM “random walk” concept failed seed stability (PM-11).

Answer to RQ1. TSFMs contain sparse causal directions that mainly control forecast level, volatility and shape, with trend and seasonality in smaller groups. They form a continuum with a few dominant axes, not discrete concepts aligned with named generative factors.

7.3 RQ2: Are concepts shared across architectures?

Geometry is shared above chance, but similarity has no single meaning. Peak CKA between TimesFM and each other model is 0.38–0.43 (null 0.037); the TimesFM/Chronos-2 peak replicates held-out (dev 0.4262; private 0.4253 [0.4162, 0.4423]), and the band holds across five architectures including Timer and Lag-Llama (SH-01). Yet seven pair metrics rank the six pairs inconsistently (Kendall’s $W = 0.28$, $p = 0.07$; Figure 7): TimesFM/Chronos-2 is first by error agreement (0.947) and fourth by CKA, so the metrics are never pooled. An untrained same-architecture pair is more CKA-similar (0.878) than a related trained pair (0.734, $p = 0.0005$; MN-01), and a unit-correspondence measure *rose* under random rotation (0.148 to 0.331) because the layers are so low-rank (SH-19).

Models agree on which inputs a concept groups. 1,223 of 1,581 atlas transfer tests are reciprocal after BH correction. Twenty claims registered from dev alone were all confirmed on the 800 private series (Holm, maximum adjusted $p = 0.0100$; AUC 0.971–0.995 against null 95th percentiles of 0.57–0.77; Figure 9b), replicating on three independent private draws (SH-18). This is an input-level claim, and the 20 claims come from two concepts with only 6 of 20 private best features identical to dev: the destination has *some* feature selecting the same series.

Models rarely agree on what a concept does. Of 27 atlas concepts, 16 have causal members from ≥ 2 models, but only 2 are *shared*, 1 partially shared, 13 *convergent* (same effect, different series) and 11 single-model (Figure 8); among the 22 seed-stable concepts the counts are 2, 1, 9 and 10. None spans all four models. Of the 1,223 reciprocal transfers,

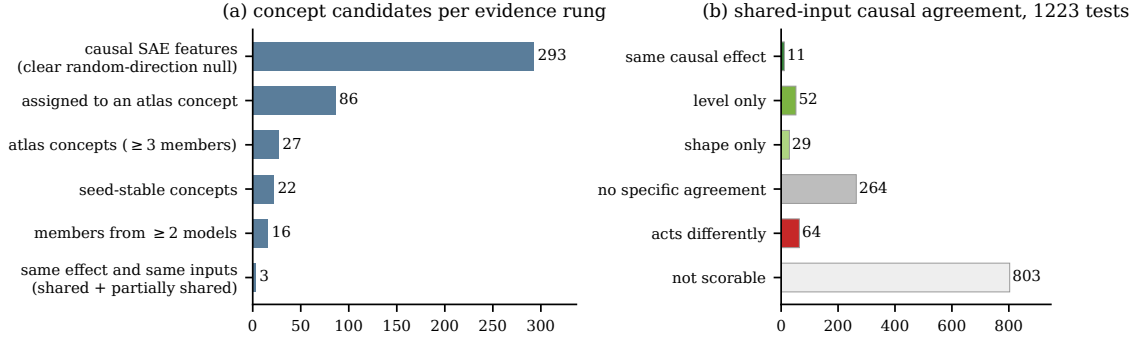


Figure 8: (a) Concept candidates at each rung (first three bars in sequence; last three are properties of the 27 atlas concepts). (b) Shared-input causal agreement over 1,223 reciprocal transfers. Source: `sae/concept_stage.json`.

420 are scorable, and only 11 (2.6%) show the same causal effect; 81 agree on level or shape alone, 264 show no specific agreement and 64 act differently.

Answer to RQ2. Different TSFMs *detect* the same things reliably enough to replicate on sealed data, but mostly do not *use* them the same way: the typical multi-model concept is convergent, and none is common to all four models. Whole-layer similarity cannot reveal this and is internally inconsistent.

7.4 RQ3: Do concepts causally affect the forecast, and how much?

Individually, little. The median single-feature effect is 0.0395 context standard deviations, 3.5 times below the 0.139 moved by merely substituting the SAE reconstruction (MN-15). Of 565 strongly activating “interesting” features, 364 (64.4%) clear no causal channel (MN-14), and TSFM-LENS collapses them rather than rendering them. Dead-atom revival improved fidelity yet lowered ground-truth alignment at every TimesFM depth tested (-0.030 to -0.132 ; MN-21). At the head level, attention mass on seasonal lags is uncorrelated with the seasonal power lost when the head is ablated ($\rho = 0.026$ TimesFM, -0.118 Chronos-T5-Base; CA-04), consistent with Jain and Wallace (2019); a minimal seasonality circuit is one head in TimesFM but five non-additive heads in Chronos-T5-Base (CA-03).

Answer to RQ3. Features and heads are causal in the controlled sense, but contributions are small and distributed, most prominent features carry nothing, and attention is an unreliable guide.

7.5 RQ4: Where in depth, and how comparable is depth?

For TimesFM/Chronos-2 all nine per-corruption fingerprint agreements replicate held-out (overall $\rho = 0.510$ [0.489, 0.527]; Figure 9a): the models react at matching depths to detrending, deseasonalizing, frequency shift and dropout, and at opposite depths to noise (-0.742), spikes (-0.780), warps (-0.658) and smoothing (-0.466). In TimesFM, whole-layer restoration is flat across depth but per-window restoration concentrates just before the horizon and grows with depth (noise: 0.068 at layer 0 to 0.412 at layer 16; DE-01). Depth

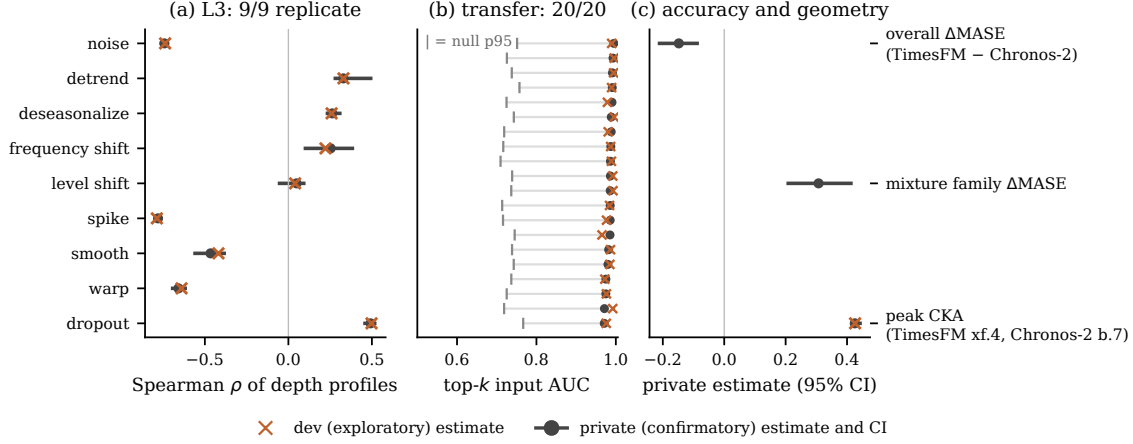


Figure 9: Claims re-tested on the 800 sealed series. (a) Per-corruption depth-profile agreement, TimesFM vs. Chronos-2; (b) the 20 concept-transfer claims with private null 95th percentiles; (c) accuracy and peak CKA. Crosses: dev; dots and bars: private with 95% series-bootstrap intervals. Source: `confirm/confirmation.json`.

locations need two qualifications that every report renders: captured compute ranges from 14% (Chronos-T5-Base) to 98% (Chronos-2) of a forecast (DE-05), and Chronos-T5’s last encoder block sits at 0.478 on the whole-stack axis (DE-04). No selected layer was robust across three layer selectors (DE-08).

Answer to RQ4. Depth-resolved reaction profiles are reproducible; absolute depth locations are comparable only after accounting for captured compute and the depth convention.

7.6 What replicated on held-out data

Of 47 registered hypotheses, 31 are currently re-testable, and every one held (Figure 9): the accuracy claim (Chronos-2 better on *mixture*, Δ MASE 0.307 [0.209, 0.413], $p_{\text{Holm}} = 0.0005$, $n = 240$; minimum detectable effect 0.155), the peak-CKA pair, all nine perturbation agreements and all 20 transfer claims (Table 5). The causal concept results of RQ1–RQ3 are exploratory, but reproduce on a second corpus: on the v1 reference run no atlas concept spans four models, convergent is again modal, and 12 of 89 scorable shared-input tests act differently (SH-16, CA-06).

8 Controls Change Conclusions

How can a reader know a system is better than a careful script running the same methods? Our answer is a record of cases in which the obvious analysis gave a well-formed, plausible and wrong answer, raised no error, and a specific control caught it. Table 6 lists fourteen; every control runs by default.

Three failure classes. *Interventions that never landed* (rows 1–4) produce flat curves or tables of zeros that read as strong negatives; the remedy (P3) is nearly free: self-patch

RQ	Result (example run unless an ID is given)	Evidence class	Status	ID
1	293 causal features; 9 families, largest act on level/volatility	causal within-model	exploratory, 2 corpora	CA-10
1	Families cover 98% but do not beat a shuffle null ($p = 1.0$)	descriptive	exploratory, 2 corpora	CA-10
1	Knob-specific concept mediation: 0/98, then 0/151	correlational	confirmed negative	MN-17
2	Peak CKA 0.425 vs null 0.037	geometric	held-out confirmed	SH-01
2	7 similarity metrics disagree ($W = 0.28$)	mixed	exploratory	SH-05/06
2	Untrained pair more similar than related trained pair	geometric	confirmed negative	MN-01
2	Concept transfer on inputs: 20/20	correlational	held-out confirmed (3 draws)	SH-18
2	No 4-model concept; convergent is the modal class	causal, compared	exploratory, 2 corpora	SH-16
2	Same effect on same inputs: 11/420 scorable	causal on shared inputs	exploratory, 2 corpora	CA-06
3	Feature effect $3.5\times$ below SAE reconstruction cost	causal within-model	exploratory	MN-15
3	64% of prominent features causally null	causal within-model	exploratory	MN-14
3	Attention mass uncorrelated with head causality	causal vs. descriptive	exploratory	CA-04
4	Depth-profile agreement, 9/9 corruptions	causal within-model	held-out confirmed	CA-01
4	Captured compute 14%–98% qualifies depth claims	descriptive	confirmed	DE-05
L0	Chronos-2 more accurate than TimesFM (-0.148)	behavioral	held-out confirmed	PM-05

Table 5: Headline results with evidence class and status. Each ID is an entry in `FINDINGS.md` with exact numbers, reproduction command and artifact path.

exactly 0.0, cross-patch nonzero, two unpatched predictions identical, front end consistent with peers. *Nulls that were too weak* (rows 5–9) produce inflated positives that look like success (30 of 30 concepts transferring); the remedy is a null matched on the right confound, on the same rows, bounded from both sides, with the lower tail for claims of difference. *Metrics that did not measure what they were named for* (rows 10–14) are caught only by validating on a known answer (P1): an untrained twin, a random rotation, known spans, a replicate seed. Smaller cases include a headline “reconstruction improves the forecast” result that replicated in 1 of 5 seeds (MN-16), and a grounded language model narrating features that produced a new class of unsupported claim after each of four guard passes, while 6 of 6 deterministic sentences were correct (MN-06); TSFM-LENS therefore composes every report sentence deterministically. The table is a set of counterexamples rather than a benchmark, and it lists only failures that were caught; the verification standard (`CONTRIBUTING.md`) requires each new test to be shown failing against a planted regression.

#	Analysis	Uncontrolled result	Controlled result	Built-in control	ID
<i>Interventions that never landed or were not identical</i>					
1	Skip lens, Chronos-2	MASE byte-identical (1.564247) at all 12 layers: “forecast formed at layer 0”	The head reads placeholder tokens the patch never writes; real reach by depth 0.737, 0.643, 0.282, 0.095, 0.000	reach probe; lens withheld when the head does not read patched positions	DE-03
2	Patching with bf16 cache	Aggregates within 0.66% of CI width of the corrected values	Self-patch was 0.00214, not 0; 71% of role assignments at one layer changed	self-patch must be exactly 0.0; clean cache with autocast off	MN-24
3	Sampled model (Sundial) as causal destination	Scorable in 1 of 71 tests: “Sundial is causally inert”	Null reseeded differently from its baseline; after the fix, scorable in 12 of 71	identical seeds across every compared pair of calls	PM-09
4	Scale equivariance, Sundial	Residual 38.43 context s.d. read as a model property; 100% flat forecasts on electricity	Adapter bypassed the checkpoint’s normalization; residual 0.030, flat share 0.08	frontend residual flagged against peers	PM-15
<i>Nulls that were too weak, or asked the wrong question</i>					
5	Concept transfer	Uniform random-subset null: 30 of 30 concepts transfer	Stratum-matched null: 56.3% forward, 46.1% reciprocal	null matched on benchmark strata	SH-14
6	Cross-model causal comparison	18 of 19 activation-matched roles “fire together, act differently”	Same series, own nulls, matched floors: 12 of 89 scorable tests act differently	shared-input agreement with matched floors	CA-06
7	Disagreement test	109 of 288 tests “act differently” (failing a q_{95} bar)	Lower-tail test: 9 act differently; 32 show no specific agreement	q_{95} required for disagreement	MN-18
8	Random-weight twins as causal floor	One floor per model	TimesFM’s twin is an exact identity stack yet reaches the forecast (0.187); Chronos twins are numerically dead ($< 10^{-7}$); floor exists for 1 of 6 pairs	within-run matched-feature floors; relative reach gate	PM-08
9	SAE names from ground truth	11 of 11 headline features named after corpus provenance	After residualizing, 82 of 88 cards (93%) ranked float-rounding noise (ρ 0.203 $\rightarrow$ 0.615)	residual-scale floor; control bounded from both sides	MN-05
<i>Metrics that did not measure what they were named for</i>					
10	CKA and stitching for lineage	Related trained pair shows high similarity (CKA 0.734)	Untrained same-architecture pair is higher (0.878; $p = 0.0005$)	random-init twin comparison	MN-01
11	Unit-level correspondence	0.24–0.54, every CI excludes 0	Rotating both bases <i>raises</i> it (0.148 $\rightarrow$ 0.331)	rotation control on real activations	SH-19
12	Captured-depth chart	Chronos-T5 fully observed (1.0), TimesFM 0.851	Share of forecast FLOPs: Chronos-T5 14.4%, TimesFM 42.5%	FLOP-based coverage from the full forecast	DE-05
13	Span refusal statistic	“Diffuse” spans: refuses 3 of 5 correct adapters	Statistic grew with token count; peak-to-pedestal contrast accepts all (IoU 1.0)	gates normalized by attainable ceiling	MP-01
14	Single-seed SAE as gold standard	Layer selector beats random on 2 of 3 architectures	Verdict flips with the extraction seed; 3-seed average agrees across seeds	replicate SAE seeds	MP-04

Table 6: Fourteen documented cases in which an uncontrolled analysis gave a well-formed wrong answer and a built-in control changed the conclusion. Every control listed runs in the default configuration. Full accounts, including the exact commands, are in `FINDINGS.md` under each ID.

9 Using and Extending TSFM-Lens

A typical session installs the two packages, builds a benchmark and runs one configuration:

```
pip install -e . && pip install -e tsfm_lens
cd tsfm_benchmark && PYTHONPATH=. python3 example_runs/run_full.py \
    --config configs/example.yaml --out ./benchmark_out --references monash
cd ../tsfm_lens && python run.py --config configs/examples/medium_2model.yaml
```

The CLI adds a preflight check (`--doctor`), partial reruns (`--stages`, `--force`), checks for new checkpoints or library upgrades (`--check-alignment`, `--check-adapter`) and an environment diff (`--verify-provenance`). A CPU-only mock configuration runs every stage in minutes in continuous integration, and the worked example (`tsfm_lens/docs/worked_example.md`) walks through a report. The four-model analysis takes 6.4 GPU-hours, mostly SAE training and the concept battery; stages through L3 take about 16 minutes. Of eight TSFM checkpoints probed, two (Timer, Time-MoE) load through the zero-code path and five with standalone packages need a hand-written adapter (MN-25), for which the template, checklist and guide exist (Appendix D). New report sections declare their artifact requirements, new claims must use the typed **Finding** record, and new configuration fields declare which artifacts they affect.

10 Limitations

Models. Four transformer models with contiguous patch tokens; disjoint-lag tokens (Lag-Llama) get reduced scope. The analysis is univariate: Chronos-2’s cross-series group attention is not measured (PM-13), and Chronos-T5’s decoder and Sundial’s flow head are not captured.

Ground truth and confirmation. Ground truth is exact only for the synthetic tier and encodes our generators’ assumptions; 27 of 76 concept parts are dominated by one real-derived generator, and pretraining overlap cannot be verified. Held-out confirmation covers accuracy, geometry, perturbation agreement and input-level transfer; the causal concept results replicate across two corpora but not yet on a sealed split, and the 20 confirmed transfers come from two concepts.

SAE dependence, usability and completeness. Concept results depend on dictionary size, sparsity, revival and seed; the system measures this dependence but cannot remove it, and feature death tracks layer geometry (MP-05). Accessibility claims rest on design; no user study was run. Controls catch only failures for which a known-answer case exists.

11 Conclusion

TSFM interpretability is harder than it looks: tokens mean different things in different models, there is no annotated ground truth, and causal effects are small enough that artifacts dominate unless controlled. TSFM-LENS addresses this with a controlled benchmark, measured alignment, null- and reach-gated causal concepts and a register-then-confirm protocol, behind one command and a report for non-specialists. On four TSFMs, causal features act mainly on forecast level and shape and form a continuum; models detect the same things, replicably, but rarely use them the same way; and individual features account

for little of the forecast. The fourteen caught failures show why these conclusions needed their controls.

Availability. Code (MIT), configurations, the example run and the findings ledger are at <https://github.com/DanK10097110/TSFM-Interp>; the `dev` branch holds the full research log and study drivers.

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Appendix A. Findings Ledger Index

Each identifier is an entry in `FINDINGS.md` giving exact numbers, models, evidence class, status, a reproduction pointer (configuration, command, artifact path) and an interestingness score (1–5); negative results are included on equal terms. Prefixes: SH shared vs. unique; CA causal agreement; DE depth; PM per-model character; MP method positives; MN method negatives; BM benchmark trust.

Table 7: Ledger entries cited in this paper (score = interestingness, 1–5).

ID	Claim (abridged)	Evidence / status	Score
SH-01	Peak CKA 0.38–0.47 above shuffled-series nulls (0.02–0.16) across five architectures, including Timer via the zero-code adapter	geometric; exploratory	5
SH-05	CKA and clustering AMI rank model pairs differently on three panels	geometric vs. descriptive; replicated	5
SH-06	Pair similarity splits into a geometric camp and a behavioral/causal camp; 7 metrics never pooled	mixed; exploratory	5
SH-14	Concept transfer on shared inputs vs. stratum-matched null: 56.3% forward, 46.1% reciprocal; uniform null gave false 30/30	correlational; exploratory	5
SH-16	No atlas concept spans all four models; convergent is the modal sharing class	descriptive over causal parts; exploratory	5
SH-18	Registered concept-transfer claims confirmed on private data: 19/20, 20/20 frozen, 20/20 on v2	held-out confirmed	5
SH-19	Unit-level correspondence fails rotation controls (0.148 $\rightarrow$ 0.331); participation ratio as low as 1.54	geometric; negative	5
SH-20	11 of 13 targets non-modular once a minimum cluster size is enforced	descriptive; confirmed	4
CA-01	Perturbation-fingerprint agreement is pair-specific; TimesFM/Chronos-2 replicates on private data	causal within-model; held-out	4
CA-03	Seasonality circuit: 1 head (TimesFM) vs. 5 non-additive heads (Chronos-T5-Base)	causal within-model; exploratory	4
CA-04	Attention mass at seasonal lags uncorrelated with causal importance ($\rho = 0.026$, -0.118)	causal vs. descriptive; exploratory	4
CA-06	On shared inputs with matched floors, most apparent cross-model disagreement disappears (18/19 $\rightarrow$ 12/89)	causal on shared inputs; exploratory, 2 corpora	5
CA-07	Median level share of a causal feature’s effect 0.535; most features shape-causal beyond a level-removed null	causal within-model; exploratory	4
CA-10	Causal effect space is a continuum; families tile it but do not beat a shuffle null	descriptive; exploratory, 2 corpora	4
DE-01	Whole-layer patching hides a depth-intensifying recency effect in the last pre-horizon window	causal within-model; exploratory	4
DE-03	Chronos-2 skip lens was a silent no-op (identical MASE at 12 layers); corrected reach 0.737 $\rightarrow$ 0.000	causal within-model; confirmed	5
DE-04	Chronos-T5 encoder ends at depth 0.478 on a whole-stack axis; 50% comparable overlap with TimesFM	descriptive; confirmed	5
DE-05	Captured share of forecast FLOPs: 14.35% (Chronos-T5-Base) to 97.5% (Chronos-2)	descriptive; confirmed	5
DE-08	Layer selection depends almost entirely on the selector (0/3 robust layers)	descriptive; confirmed	4
PM-05	Chronos-2 most accurate and best observed; winners vary by family; held-out on v2	behavioral; partly held-out	4
PM-07	Models differ $\sim 5.4\times$ in reversion to the mean; flat forecasts driven by weakly predictable data	behavioral; confirmed on run	4

ID	Claim (abridged)	Evidence / status	Score
PM-08	Random-weight twins are not one null: identity stack (TimesFM) vs. dead forecasts (Chronos-2/Bolt)	architecture fact; confirmed	5
PM-09	Sundial scorable as causal destination 1/71 $\rightarrow$ 12/71 after seeding fix	causal; corrected	3
PM-11	Causally load-bearing TimesFM random-walk concept fails seed stability	causal within-model; unstable	3
PM-13	Chronos-2 group attention unmeasured by design	scope statement	3
PM-15	Sundial adapter bypassed checkpoint normalization; residual 38.43 $\rightarrow$ 0.030	behavioral; fixed	4
MP-01	Token spans can be measured (IoU 1.0 on 6 checkpoints); zero-code adapter follows	method; confirmed	5
MP-02	Alignment gates must be normalized by their attainable ceiling	method; confirmed	3
MP-03	AuxK revival recovers dead SAE atoms 10–13 $\times$ on real activations	method; adopted	4
MP-04	Single-seed SAE gold ranking flips a bake-off verdict; 3 seeds fix it	method; fixed	5
MP-05	SAE feature death set by layer geometry (exponent 1.02), not recipe	correlational; replicated	5
MP-06	Ground-truth alignment of SAE features is real only over a permutation null	correlational; exploratory	3
MP-07	Ablation battery discriminates: 2.49–6.61 $\times$ chance	causal within-model; confirmed	4
MP-09	Dev and private splits exchangeable (energy $p = 0.51$; TOST 21/22)	held-out validation	3
MN-01	Similarity cannot detect lineage: untrained pair 0.878 > trained related pair 0.734	geometric; negative	5
MN-05	Residualizing on provenance made 93% of headline cards float-rounding noise	method; fixed	5
MN-06	A grounded small LLM narrator fabricates comparisons; guards do not converge	method; negative	5
MN-07	Causal fingerprints are nearly one-dimensional (PR 2.09)	causal, compared; negative	4
MN-14	64.4% of visually prominent SAE features are causally null	causal within-model	4
MN-15	SAE reconstruction moves the forecast $\sim 3.5\times$ more than removing a feature	causal within-model	4
MN-16	“Reconstruction beats clean forecast” replicated in 1/5 seeds; retracted	causal; retracted	4
MN-17	Generator-counterfactual concept mediation: 0/98 and 0/151	correlational; negative	5
MN-18	Absence of agreement is not disagreement: 109/288 $\rightarrow$ 9	method; fixed	4
MN-19	“Same inputs” must be top- k overlap, not whole-series correlation	method; fixed	4
MN-21	Reviving dead features hurts ground-truth alignment in TimesFM at every depth	correlational vs. causal; exploratory	5
MN-23	MASE denominator floors out on intermittent series	metric caveat	3
MN-24	A precision mismatch left aggregates intact but reorganized 71% of a partition	method; fixed	4
MN-25	Zero-code adapter resolves 2 of 8 probed TSFM checkpoints	descriptive; open	3
BM-05	Counterfactual identity check caught a corruption-replay bug (244/298 $\rightarrow$ 0/1113)	benchmark; fixed	2

Appendix B. Statistical Details

Paired series bootstrap. With per-series differences $d_s = m_{a,s} - m_{b,s}$ and $B = n_{\text{boot}}$ resampled means $\bar{d}_b^*$, the interval is the percentile interval and $p = \min(1, \max(\frac{1}{B}, 2 \min\{\frac{1}{B} \#(\bar{d}_b^* \leq 0), \frac{1}{B} \#(\bar{d}_b^* \geq 0)\}))$ (`analysis/stats.py`; floor per Phipson and Smyth, 2010). Window-level statistics (CKA, fingerprints) resample series with all their windows (Cameron et al., 2008).

Multiplicity. Holm: $\tilde{p}_{(i)} = \max_{j \leq i} \min(1, (m - j + 1)p_{(j)})$ (Holm, 1979); with $p \geq 1/B$ the smallest attainable value is m/B , which the ledger reports per family, flagging $m/B > \alpha$. BH: $\tilde{q}_{(i)} = \min_{j \geq i} \min(1, \frac{m}{j} p_{(j)})$ (Benjamini and Hochberg, 1995).

Row-matched ablation null. For feature i with rows R_i and J unit directions u_j , the null removes $\bar{a} W_d u_j$ ($\bar{a}$: mean activation removed by the batch’s ablations); channel c clears if $\frac{1}{|R_i|} \sum_{r \in R_i} |\Delta_{i,r}^{(c)}| > q_{0.95} \{|\Delta_{u_j,r}^{(c)}| : j \leq J, r \in R_i\}$. A null with no spread is reported as degenerate and not scored.

Transfer. The statistic is $\max_f \text{AUC}(a_f; S)$ over destination features; its null recomputes the same maximum on stratum-matched random k -sets, so the feature search is inside the null. The reverse leg scores the chosen feature’s own top- k by the source activation. $p = (1 + \#\{\text{null} \geq \text{obs}\}) / (1 + n)$; a registered claim uses $\max(p, p_{\text{rev}})$ (intersection–union), Holm across claims.

Shared-input agreement. On U , statistic (i) is the Spearman concordance of the two sides’ signed level effects across series, and (ii) the cosine of their level-removed shape vectors, each channel in its own null units. Each side’s floor ablates random feature sets matched to its activation on U and re-scores against the other side’s measured effect; agreement exceeds both floors’ $q_{0.95}$, “acts differently” falls below both $q_{0.05}$.

Minimum detectable effect. Before the private result is read, recentred per-series differences are shifted by a candidate effect and re-tested with the same bootstrap; bisection finds the smallest shift detected with power 0.8 (`analysis/power.py`).

Appendix C. Reproducibility

`examples/concept_atlas_v2` holds the resolved configuration, the report and every stage’s JSON/parquet artifacts (activations and SAE weights omitted for size); `paper/figures/make_figures.py --run DIR` regenerates all data figures. From `tsfm_lens/`, `python run.py --config configs/concept_atlas_v2.yaml` reproduces the run (add `--stages report` to re-render, and `--verify-provenance runs/concept_atlas_v2` to diff environments). The private split was opened once (2026-09-28), so re-confirming requires a fresh epoch. Environment: Python 3.12, numpy 2.1.0, PyTorch 2.9.1 (CUDA 13.0), transformers 4.57.6, timesfm 2.0.2, chronos-forecasting 2.3.1, zarr 2.18.7 (zarr 3 stores read back empty under the v2 API), scikit-learn 1.9.0; all pins with reasons are in `DEPENDENCIES.md`. Tests: 2,112 passed, 27 skipped (`tsfm_lens`); 124 passed, 1 skipped (`tsfm_benchmark`). Wall-clock time on one RTX A5000: 23,151 s in total, of which `sae` 8,594 s and `concepts` 13,377 s (shared-input agreement 10,855 s); every other stage under 4 minutes.

Appendix D. Adapter Pitfalls: a Worked Case

Sundial z-scores inputs inside `generate()`, but `forward()` defaults to no normalization. `generate()` was broken on the installed transformers, so the adapter called `forward()` and fed raw-scale input (PM-15). The `frontend` stage measured a scale-equivariance residual of 38.43 context standard deviations (peers ≤ 0.009) and 100% flat forecasts on electricity, and the residual was first recorded as a model property. The adapter now normalizes in `prepare()` with the checkpoint’s exact rule and denormalizes in `predict()`, so capture, prediction and patching see identical inputs: residual 0.030, flat share 0.08; the same defect in `generic_hf` on Timer fell from 49.73 to 0.020. The report now flags any

residual above $\max(1, 10 \times \text{the peers' median})$. Lesson: an adapter calling a model below its public API must reproduce everything that API does to input and output. The guide (`tsfm_lens/docs/ADDING_A_MODEL.md`) turns this and the other pitfalls into a checklist: normalization, identical seeds for compared calls of sampled heads, clean-cache precision, no batch chunking, anchored layer regexes, measured contiguous spans, stated encoder–decoder capture, reach controls, twin reach checks, original-scale outputs, and re-checking alignment after any library or checkpoint change.

References

- J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, 2018.
- G. Alain and Y. Bengio. Understanding intermediate layers using linear classifier probes. In *International Conference on Learning Representations (Workshop Track)*, 2017.
- A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024.
- A. F. Ansari, O. Shchur, et al. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025.
- Y. Bansal, P. Nakkiran, and B. Barak. Revisiting model stitching to compare neural representations. In *Advances in Neural Information Processing Systems*, 2021.
- Y. Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.
- N. Belrose, Z. Furman, L. Smith, D. Halawi, I. Ostrovsky, L. McKinney, S. Biderman, and J. Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.
- Y. Benjamini and Y. Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1): 289–300, 1995.
- J. Bloom, C. Tigges, A. Duong, and D. Chanin. SAELens. <https://github.com/jbloomAus/SAELens>, 2024.
- T. Bricken, A. Templeton, J. Batson, B. Chen, A. Jermyn, T. Conerly, N. Turner, C. Anil, C. Denison, A. Askell, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
- A. C. Cameron, J. B. Gelbach, and D. L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008.

- A. Das, W. Kong, R. Sen, and Y. Zhou. A decoder-only foundation model for time-series forecasting. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- F. Divo, M. Kraus, R. Härle, P. Ahrend, and K. Kersting. Finding the Zeitgeist in time series foundation models. In *ICLR 2026 Workshop on Time Series in the Age of Large Models*, 2026.
- C. Dwork, V. Feldman, M. Hardt, T. Pitassi, O. Reingold, and A. Roth. The reusable holdout: Preserving validity in adaptive data analysis. *Science*, 349(6248):636–638, 2015.
- B. Efron and R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1993.
- J. Fiotto-Kaufman, A. R. Loftus, E. Todd, J. Brinkmann, K. Pal, D. Troitskii, M. Ripa, A. Belfki, C. Rager, C. Juang, et al. NNSight and NDIF: Democratizing access to open-weight foundation model internals. In *International Conference on Learning Representations*, 2025.
- L. Gao, T. Dupré la Tour, H. Tillman, G. Goh, R. Troll, A. Radford, I. Sutskever, J. Leike, and J. Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, 2025.
- P. Gao, E. Trautmann, B. Yu, G. Santhanam, S. Ryu, K. Shenoy, and S. Ganguli. A theory of multineuronal dimensionality, dynamics and measurement. *bioRxiv*, page 214262, 2017.
- T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- R. Godahewa, C. Bergmeir, G. I. Webb, R. J. Hyndman, and P. Montero-Manso. Monash time series forecasting archive. In *Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.
- M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski. MOMENT: A family of open time-series foundation models. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.
- S. Heimersheim and N. Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.
- J. Hewitt and P. Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019.
- S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.
- M. Huh, B. Cheung, T. Wang, and P. Isola. Position: The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.
- R. J. Hyndman and A. B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006.

- A. A. Ismail, M. Gunady, H. Corrada Bravo, and S. Feizi. Benchmarking deep learning interpretability in time series predictions. In *Advances in Neural Information Processing Systems*, 2020.
- S. Jain and B. C. Wallace. Attention is not explanation. In *Proceedings of NAACL-HLT*, 2019.
- M. Kalnāre, S. Kitharidis, T. Bäck, and N. van Stein. Mechanistic interpretability for transformer-based time series classification. *arXiv preprint arXiv:2511.21514*, 2025.
- S. Kapoor and A. Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 2023.
- T. Kim, J. Kim, Y. Tae, C. Park, J.-H. Choi, and J. Choo. Reversible instance normalization for accurate time-series forecasting against distribution shift. In *International Conference on Learning Representations*, 2022.
- N. Kokhlikyan, V. Miglani, M. Martin, E. Wang, B. Alsallakh, J. Reynolds, A. Melnikov, N. Kliushkina, C. Araya, S. Yan, and O. Reblitz-Richardson. Captum: A unified and generic model interpretability library for PyTorch. *arXiv preprint arXiv:2009.07896*, 2020.
- S. Kornblith, M. Norouzi, H. Lee, and G. Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, 2019.
- H. R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241, 1989.
- Y. Li, J. Yosinski, J. Clune, H. Lipson, and J. Hopcroft. Convergent learning: Do different neural networks learn the same representations? In *International Conference on Learning Representations*, 2016.
- J. Lindsey, A. Templeton, J. Marcus, T. Conerly, J. Batson, and C. Olah. Sparse crosscoders for cross-layer features and model diffing. *Transformer Circuits Thread*, 2024.
- Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- Y. Liu, H. Zhang, C. Li, X. Huang, J. Wang, and M. Long. Timer: Generative pre-trained transformers are large time series models. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.
- Y. Liu, G. Qin, Z. Shi, Z. Chen, C. Yang, X. Huang, J. Wang, and M. Long. Sundial: A family of highly capable time series foundation models. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. arXiv:2502.00816.
- C. H. Lubba, S. S. Sethi, P. Knaute, S. R. Schultz, B. D. Fulcher, and N. S. Jones. catch22: CAnonical time-series characteristics. *Data Mining and Knowledge Discovery*, 33(6): 1821–1852, 2019.

- A. Makelov, G. Lange, A. Geiger, and N. Nanda. Is this the subspace you are looking for? an interpretability illusion for subspace activation patching. In *International Conference on Learning Representations*, 2024.
- A. Makhzani and B. Frey. k-sparse autoencoders. *arXiv preprint arXiv:1312.5663*, 2013.
- K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022.
- A. Mishra. Dissecting Chronos: Sparse autoencoders reveal causal feature hierarchies in time series foundation models. In *ICLR 2026 Workshop on Time Series in the Age of Large Models*, 2026. arXiv:2603.10071.
- N. Nanda and J. Bloom. TransformerLens. <https://github.com/TransformerLensOrg/TransformerLens>, 2022.
- B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018.
- K. Oublal, Q. Bouniot, Q. Gan, S. Cl  men  on, and Z. Akata. TimeSAE: Causal sparse decoding for faithful explanations of black-box time series models. *arXiv preprint arXiv:2601.09776*, 2026.
- G. Paulo and N. Belrose. Sparse autoencoders trained on the same data learn different features. *arXiv preprint arXiv:2501.16615*, 2025.
- B. Phipson and G. K. Smyth. Permutation p-values should never be zero: Calculating exact p-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1), 2010.
- C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.
- M. Raghu, J. Gilmer, J. Yosinski, and J. Sohl-Dickstein. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems*, 2017.
- K. Rasul, A. Ashok, A. R. Williams, H. Ghonia, R. Bhagwatkar, A. Khorasani, M. J. Darvishi Bayazi, G. Adamopoulos, R. Riachi, N. Hassen, M. Bilo  , S. Garg, A. Schneider, N. Chapados, A. Drouin, V. Zantedeschi, Y. Nevmyvaka, and I. Rish. Lag-Llama: Towards foundation models for probabilistic time series forecasting. *arXiv preprint arXiv:2310.08278*, 2023.
- P. J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.
- H. Sakoe and S. Chiba. Dynamic programming algorithm optimization for spoken word recognition. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 26(1):43–49, 1978.

- D. J. Schuurmann. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6):657–680, 1987.
- X. Shi, S. Wang, Y. Nie, D. Li, Z. Ye, Q. Wen, and M. Jin. Time-MoE: Billion-scale time series foundation models with mixture of experts. In *International Conference on Learning Representations*, 2025.
- G. J. Székely and M. L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013.
- J. Vig, S. Gehrmann, Y. Belinkov, S. Qian, D. Nevo, Y. Singer, and S. Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, 2020.
- N. X. Vinh, J. Epps, and J. Bailey. Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance. *Journal of Machine Learning Research*, 11:2837–2854, 2010.
- S. Wan, M. Ma, H. Zhu, G. Liu, S. Wang, Q. Wen, and H. Liu. Forecast collapse in time-series foundation models. *arXiv preprint arXiv:2608.14106*, 2026.
- M. Wiliński, M. Goswami, W. Potosnak, N. Żukowska, and A. Dubrawski. Exploring representations and interventions in time series foundation models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025.
- G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo. Unified training of universal time series forecasting transformers. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.
- Z. Wu, A. Geiger, A. Arora, J. Huang, Z. Wang, N. D. Goodman, C. D. Manning, and C. Potts. pyvene: A library for understanding and improving PyTorch models via interventions. In *Proceedings of NAACL-HLT (System Demonstrations)*, 2024.
- A. Yıldırım. Superposition is not necessary: A mechanistic interpretability analysis of transformer representations for time series forecasting. *arXiv preprint arXiv:2605.05151*, 2026.
- F. Zhang and N. Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations*, 2024.
- H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, 2021.